

Technical Brief: Multi-Agent Market Intelligence Ecosystem

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Stack: OpenClaw, Python, SQLite, Gemini 1.5 Pro, local Gemma (27b/4b)

1. Executive Summary

A production-ready autonomous research agent designed to solve the "hallucination vs. latency" tradeoff in enterprise data extraction. The system utilizes a multi-agent orchestration strategy to crawl, parse, and store market intelligence with durable memory and human-verified approval gates.

2. Architectural Framework: Hybrid Inference Routing

To optimize for both deep reasoning and operational cost/speed, the system implements a **tri-tier routing strategy**:

- **Tier 1 (Reasoning):** Complex planning and cross-referencing tasks are routed to **Gemini 1.5 Pro**.
- **Tier 2 (Verification):** Local **Gemma 27b** handles data validation and consistency checks, keeping sensitive validation logic on-premise.
- **Tier 3 (Formatting):** High-velocity, low-complexity parsing and JSON schema formatting are handled by **Gemma 4b** to minimize latency.

3. Key Innovation: Durable Memory & Autonomous ETL

Unlike standard "stateless" chat agents, this system features:

- **Durable State Management:** Backed by a **SQLite** persistence layer, ensuring that agents retain context across long-running research tasks even if a network connection is interrupted.
- **Autonomous ETL Pipelines:** Automated Python workers that transform unstructured web data into structured relational tables without manual intervention, utilizing HITL (Human-in-the-Loop) terminal approvals for high-stakes data commits.

4. Applied Value for Research Hubs

This architecture is directly applicable to high-level research goals, such as those of the **Microsoft Discovery** platform:

- **Scientific Discovery:** Orchestrating agents to scan massive volumes of research papers for specific variable correlations.
- **Governance & Privacy:** Local model integration (Gemma) allows for sensitive data processing without exposing PII to external APIs—a critical requirement for state-level

and academic partnerships.